

SDMTS Supply Chain Mapping Project

One-page portfolio brief

At a Glance

Client	South Dakota Manufacturing and Technology Solutions (SDMTS)
Context	Supply Chain Optimization and Intelligence Network (SCOIN) support
Primary question	How can South Dakota manufacturing records be cleaned, mapped, filtered, and interpreted for supply-chain and economic-development use?
Primary deliverables	Interactive mapping, Excel dashboard workflow, data validation logic, and gap-analysis interpretation
Public portfolio treatment	Redacted/synthetic workbook demo plus summary report, methods note, and visuals

Project Summary

This project helped transform a manufacturing company list into a more usable decision-support tool. The team worked through cleanup, validation, classification, geocoding, mapping, dashboard design, and interpretation. The project emphasized practical usability for SDMTS and South Dakota manufacturers rather than purely academic modeling.

Role Note

This was a team consulting project with shared technical and client-facing responsibilities. My primary contributions centered on interpreting the analysis, developing the written and presentation deliverables, and translating the findings into client-facing recommendations.

Technical Components

- Record cleanup and deduplication
- Validity/confidence scoring using phone, URL, D-U-N-S, address, and active-status signals
- Geocoding and map-readiness review
- Three-digit NAICS grouping for map classification
- Excel dashboard with filter/search/pivot logic
- BatchGeo and Excel Power Map comparison
- Gap-analysis framing for economic-development use

Why It Matters

Supply-chain mapping can help economic-development organizations see where manufacturers are concentrated, where industry-specific supplier gaps may exist, and where targeted outreach, infrastructure planning, or business attraction efforts may be worth exploring.